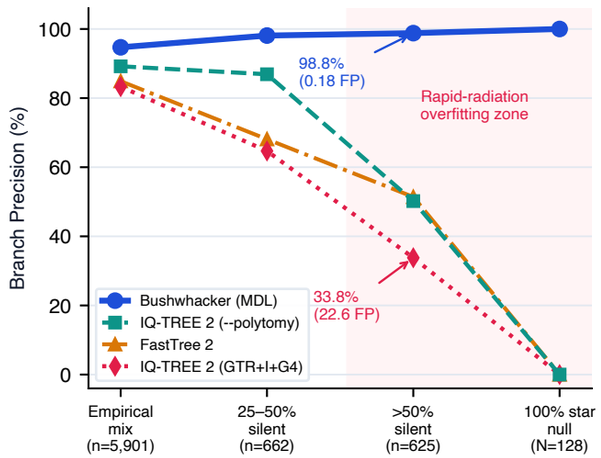
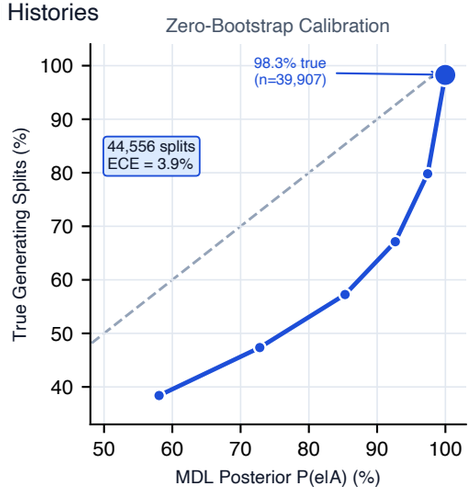
A Ground-Truth Validation Across 36,920 Simulated Histories



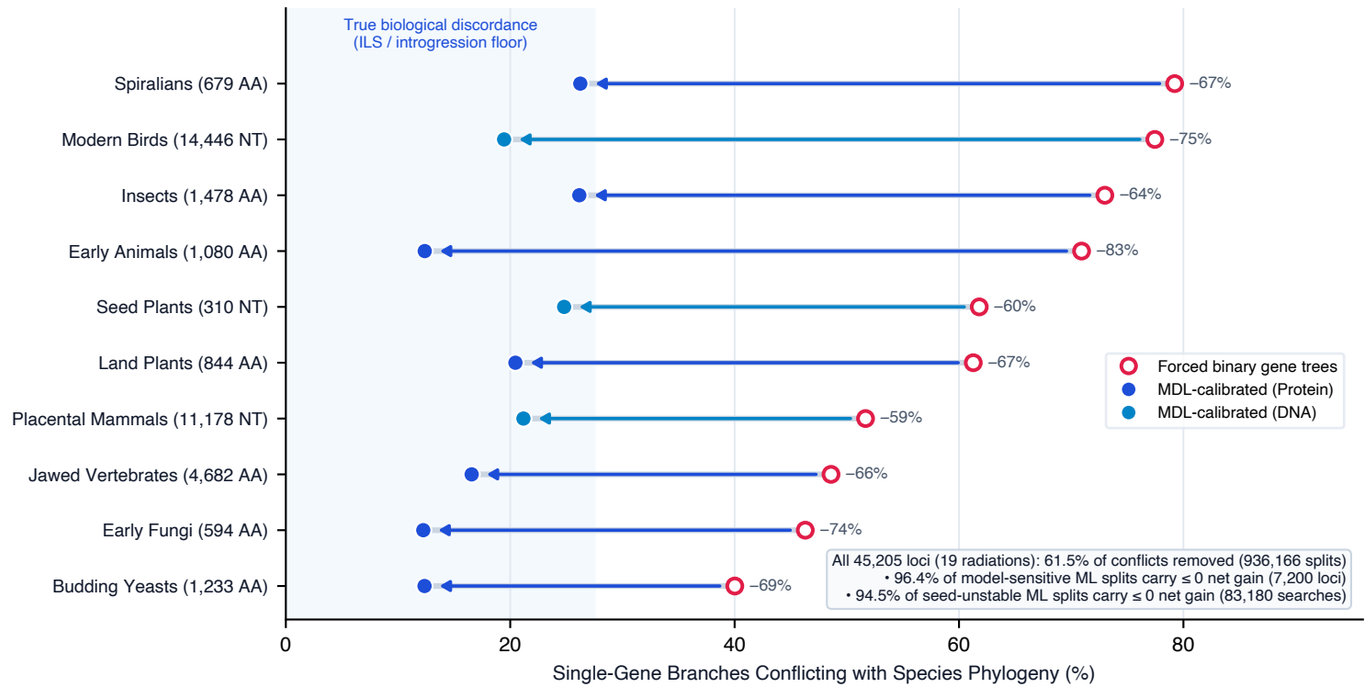

DNA & Coalescent (35,720)
92.4–98.8% prec
+16% Anomaly Zone

Protein LG+G4 (400)
99.8% rad. prec
0.02 star FP/tree

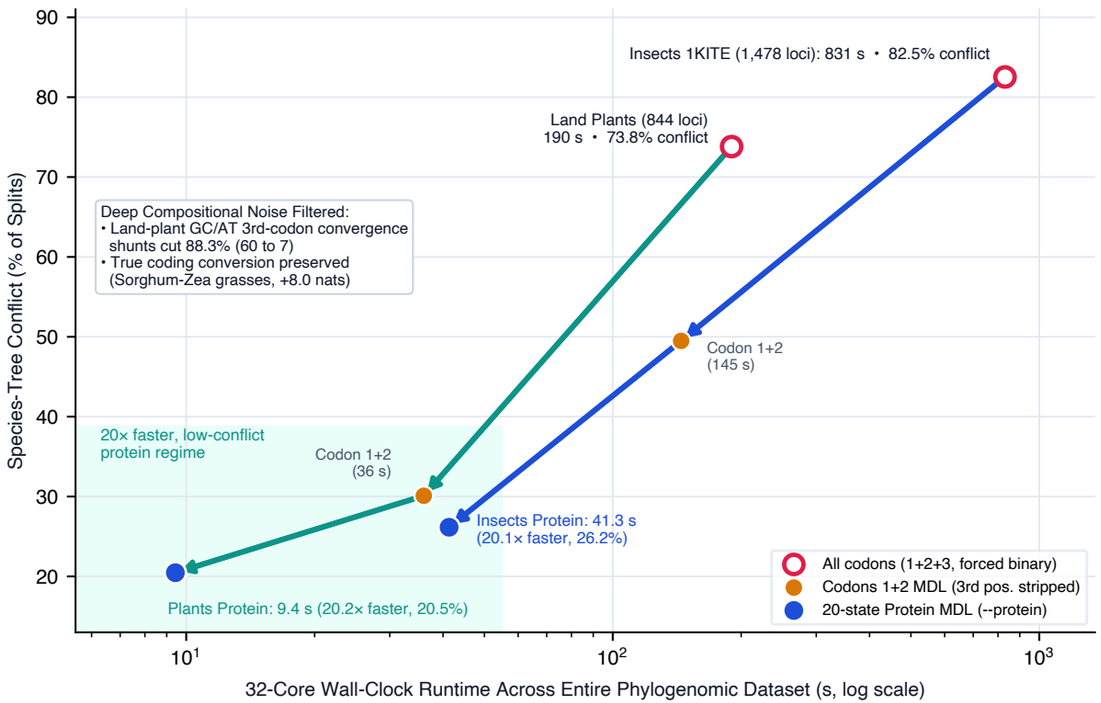
Lateral Networks (400)
91% HGT / 100% Conv
0% clonal FPR

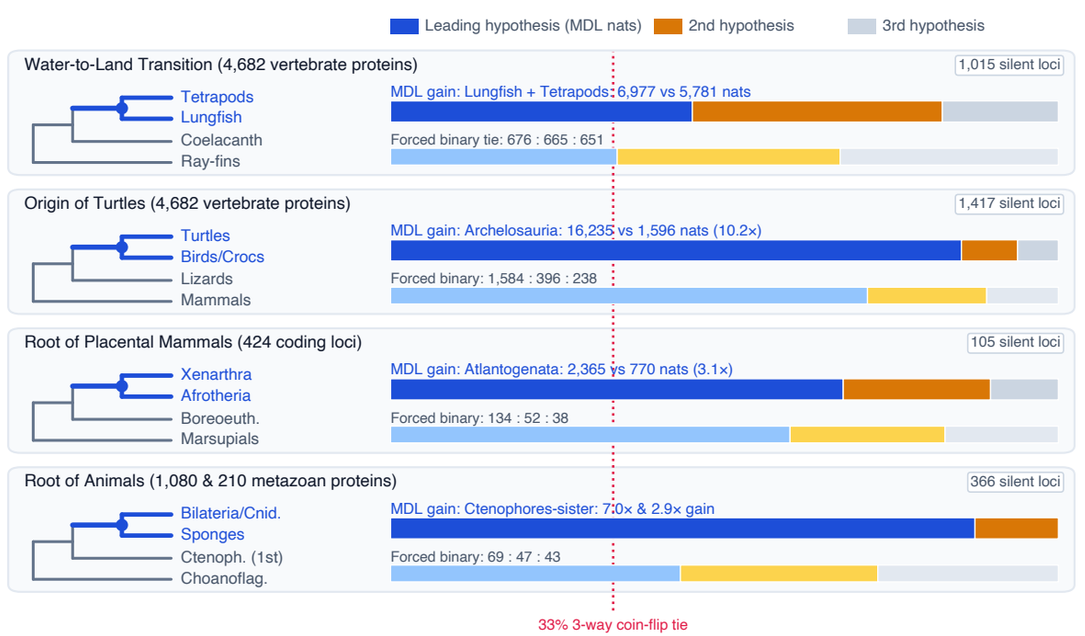
Single-Cell 0/1 & CNA (400)
86.2% binary SNV
89.9% ordered CNA

B Silent Radiations Account for Most Apparent Gene-Tree Conflict




C Protein Compression Strips Synonymous Saturation 20× Faster




D Removing Silent Coin-Flip Votes Resolves Deep Evolutionary Debates